

```
1 grammar = {'S': [['NP', 'VP']], 'NP': [['Det', 'N']], 'VP': [['V', 'NP']], 'Det': [['the'], ['a']], 'N': [['cat'], ['dog']], 'V': [['sees'], ['likes']]}
2 sentence = input("Enter sentence: ").lower().split()
3 def earley_parse(words): 1 usage
4     chart = [[] for _ in range(len(words) + 1)]
5     chart[0].append(('S', ['NP', 'VP'], 0, 0))
6     for i in range(len(words) + 1):
7         changed = True
8         while changed:
9             changed = False
10            for lhs, rhs, dot, start in chart[i]:
11                if dot < len(rhs):
12                    symbol = rhs[dot]
13                    # Prediction
14                    if symbol in grammar:
15                        for rule in grammar[symbol]:
16                            state = (symbol, rule, 0, i)
17                            if state not in chart[i]:
18                                chart[i].append(state)
19                                changed = True
20                    # Scanning
21                    elif i < len(words) and symbol == words[i]:
22                        state = (lhs, rhs, dot + 1, start)
23                        if state not in chart[i + 1]:
24                            chart[i + 1].append(state)
25            else:
26                # Completion
27                for l2, r2, d2, s2 in chart[start]:
28                    if d2 < len(r2) and r2[d2] == lhs:
29                        state = (l2, r2, d2 + 1, s2)
30                        if state not in chart[i]:
31                            chart[i].append(state)
```



```
3 def earley_parse(words): 1 usage
20     # Scanning
21     elif i < len(words) and symbol == words[i]:
22         state = (lhs, rhs, dot + 1, start)
23         if state not in chart[i + 1]:
24             chart[i + 1].append(state)
25     else:
26         # Completion
27         for l2, r2, d2, s2 in chart[start]:
28             if d2 < len(r2) and r2[d2] == lhs:
29                 state = (l2, r2, d2 + 1, s2)
30                 if state not in chart[i]:
31                     chart[i].append(state)
32                     changed = True
33     return ('S', ['NP', 'VP'], 2, 0) in chart[len(words)]
34 if earley_parse(sentence):
35     print("Sentence is accepted")
36 else:
37     print("Sentence is rejected")
```

Run main

/Users/moshika/PyCharmMiscProject/.venv/bin/python /Users/moshika/PyCharmMiscProject/Exp 12.py

Enter sentence: the cat sees a dog

Sentence is accepted

Process finished with exit code 0

|

